DSA
Practical No. 1
Arrays

Name: Arjun Mankuskar

PRN:202401110038

Batch:A2

Movie Ticket Booking System

Code:-

```
#include <iostream>
using namespace std;

const int row = 6;
const int col = 6;

// Function declarations
void showmore();
void book_seat(int movie[row][col]);
void cancel(int movie[row][col]);
void display(int movie[row][col]);
void countticket(int movie[row][col]);
void totalprice(int movie[row][col]);

// Menu display
void showmore() {
    cout << "\nMovie Booking System" << endl;
    cout << "1. Book Ticket" << endl;
```

```
    cout << "2. Cancel Ticket" << endl;
    cout << "3. Display Seats" << endl;
    cout << "4. Count Booked Seats" << endl;
    cout << "5. Total Price" << endl;
    cout << "6. Exit" << endl;
}

// Book a seat
void book_seat(int movie[row][col]) {
    int r, c;
    cout << "Enter row and column for seat booking: ";
    cin >> r >> c;

    if (r >= 0 && r < row && c >= 0 && c < col) {
        if (movie[r][c] == 0) {
            movie[r][c] = 1;
            string seatType;
            int price = 0;

            if (r == 0 || r == 1) {
                seatType = "Diamond";
                price = 500;
```

```
            } else if (r == 2 || r == 3) {
                seatType = "Gold";
                price = 350;
            } else {
                seatType = "Silver";
                price = 200;
            }

            cout << "You have selected a " << seatType << " seat.
Price: Rs. " << price << endl;
            cout << "Your seat is booked!" << endl;
        } else {
            cout << "Sorry, that seat is already booked!" << endl;
        }
    } else {
        cout << "Invalid seat position!" << endl;
    }
}

// Cancel a booking
void cancel(int movie[row][col]) {
    int r, c;
```

```
    cout << "Enter row and column to cancel booking: ";
    cin >> r >> c;

    if (r >= 0 && r < row && c >= 0 && c < col) {
        if (movie[r][c] == 1) {
            movie[r][c] = 0;
            cout << "Your booking has been canceled." << endl;
        } else {
            cout << "That seat was not booked." << endl;
        }
    } else {
        cout << "Invalid seat position!" << endl;
    }
}

// Display seat layout
void display(int movie[row][col]) {
    cout << "\nSeat Layout (0 = Available, 1 = Booked):" << endl;
    for (int i = 0; i < row; i++) {
        for (int j = 0; j < col; j++) {
            cout << movie[i][j] << " ";
```

```
        }
        cout << endl;
    }
}

// Count booked seats
void countticket(int movie[row][col]) {
    int count = 0;
    for (int i = 0; i < row; i++) {
        for (int j = 0; j < col; j++) {
            if (movie[i][j] == 1) {
                count++;
            }
        }
    }
    cout << "Total booked seats: " << count << endl;
}

// Calculate total price
void totalprice(int movie[row][col]) {
    int total = 0;
    for (int i = 0; i < row; i++) {
```

```
        for (int j = 0; j < col; j++) {
            if (movie[i][j] == 1) {
                if (i == 0 || i == 1) {
                    total += 500; // Diamond
                } else if (i == 2 || i == 3) {
                    total += 350; // Gold
                } else {
                    total += 200; // Silver
                }
            }
        }
    }
    cout << "Total price: Rs. " << total << endl;
    cout << "Total price with GST (18%): Rs. " << (0.18 * total + total) << endl;
}

// Main function
int main() {
    int choice;
    int movie[row][col] = {0};
```

```
do {
    showmore();
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
        case 1:
            book_seat(movie);
            break;
        case 2:
            cancel(movie);
            break;
        case 3:
            display(movie);
            break;
        case 4:
            countticket(movie);
            break;
        case 5:
            totalprice(movie);
            break;
        case 6:
```

```
            cout << "Thank you! Visit again." << endl;
            break;
        default:
            cout << "Wrong choice. Please try again." << endl;
    }
    cout << endl;
} while (choice != 6);

return 0;
}
```

Output:-

/*

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 1

Enter row and column for seat booking: 1

1

You have selected a Diamond seat. Price: Rs. 500

Your seat is booked!

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 1

Enter row and column for seat booking: 2

1

You have selected a Gold seat. Price: Rs. 350

Your seat is booked!

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 1

Enter row and column for seat booking: 3

1

You have selected a Gold seat. Price: Rs. 350

Your seat is booked!

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 1

Enter row and column for seat booking: 3

2

You have selected a Gold seat. Price: Rs. 350

Your seat is booked!

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 2

Enter row and column to cancel booking: 3

2

Your booking has been canceled.

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 3

Seat Layout (0 = Available, 1 = Booked):

0 0 0 0 0 0

0 1 0 0 0 0

0 1 0 0 0 0

0 1 0 0 0 0

0 0 0 0 0 0

0 0 0 0 0 0

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 4

Total booked seats: 3

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 5

Total price: Rs. 1200

Total price with GST (18%): Rs. 1416

Movie Booking System

1. Book Ticket

2. Cancel Ticket

3. Display Seats

4. Count Booked Seats

5. Total Price

6. Exit

Enter your choice: 6

Thank you! Visit again.

*/

Student Mark Analysis System

Code:-

```
#include <iostream>
using namespace std;

const int max_students = 6;

// Menu display
void showmore() {
    cout << "\nStudent Mark Analysis System" << endl;
    cout << "1. Enter Marks" << endl;
    cout << "2. Delete Record" << endl;
    cout << "3. Display All Marks" << endl;
    cout << "4. Count Students" << endl;
    cout << "5. Average Marks" << endl;
    cout << "6. Exit" << endl;
}

// Enter student marks
```

```
void enter_marks(string names[], int marks[], int &count) {
    if (count >= max_students) {
        cout << "Maximum student limit reached!" << endl;
        return;
    }

    cout << "Enter student name: ";
    cin >> names[count];
    cout << "Enter marks: ";
    cin >> marks[count];

    cout << "Record added successfully!" << endl;
    count++;
}

// Delete student record
void delete_record(string names[], int marks[], int &count) {
    string name;
    cout << "Enter student name to delete: ";
    cin >> name;

    bool found = false;
```

```
    for (int i = 0; i < count; i++) {
        if (names[i] == name) {
            for (int j = i; j < count - 1; j++) {
                names[j] = names[j + 1];
                marks[j] = marks[j + 1];
            }
            count--;
            found = true;
            cout << "Record deleted successfully!" << endl;
            break;
        }
    }

    if (!found) {
        cout << "Student not found!" << endl;
    }
}

// Display all marks
void display_marks(string names[], int marks[], int count) {
    cout << "\nStudent Marks:" << endl;
    for (int i = 0; i < count; i++) {
```

```
        cout << "Name: " << names[i] << " | Marks: " << marks[i]
<< endl;
    }
    if (count == 0) {
        cout << "No records available." << endl;
    }
}

// Count students
void count_students(int count) {
    cout << "Total number of students: " << count << endl;
}

// Calculate average marks
void average_marks(int marks[], int count) {
    if (count == 0) {
        cout << "No marks to calculate average." << endl;
        return;
    }

    int total = 0;
    for (int i = 0; i < count; i++) {
```

```
        total += marks[i];
    }
    float average = static_cast<float>(total) / count;
    cout << "Average marks: " << average << endl;
}

// Main function
int main() {
    int choice;
    string names[max_students];
    int marks[max_students] = {0};
    int count = 0;

    do {
        showmore();
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                enter_marks(names, marks, count);
                break;
```

```
            case 2:
                delete_record(names, marks, count);
                break;
            case 3:
                display_marks(names, marks, count);
                break;
            case 4:
                count_students(count);
                break;
            case 5:
                average_marks(marks, count);
                break;
            case 6:
                cout << "Thank you! Session ended." << endl;
                break;
            default:
                cout << "Wrong choice. Please try again." << endl;
        }
        cout << endl;
    } while (choice != 6);
    return 0;
}
```

Output:-

/*

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 1

Enter student name: Arjun

Enter marks: 254

Record added successfully!

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 1

Enter student name: sujal

Enter marks: 142

Record added successfully!

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 1

Enter student name: om

Enter marks: 123

Record added successfully!

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 2

Enter student name to delete: om

Record deleted successfully!

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 3

Student Marks:

Name: Arjun | Marks: 254

Name: sujal | Marks: 142

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 4

Total number of students: 2

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 5

Average marks: 198

Student Mark Analysis System

1. Enter Marks

2. Delete Record

3. Display All Marks

4. Count Students

5. Average Marks

6. Exit

Enter your choice: 6

Thank you! Session ended.

*/